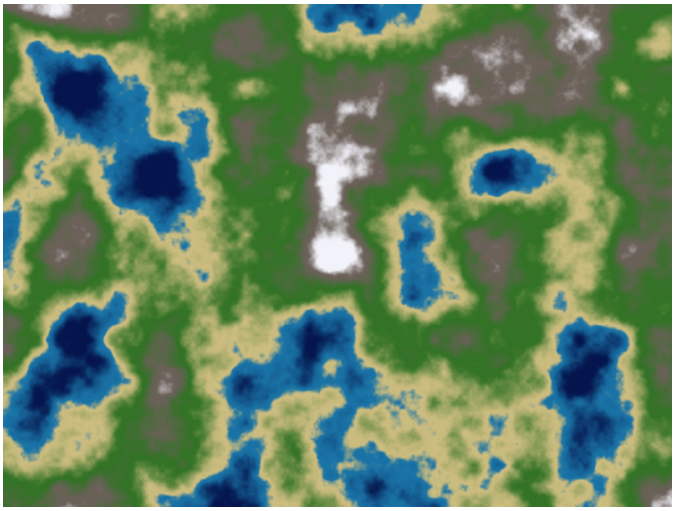
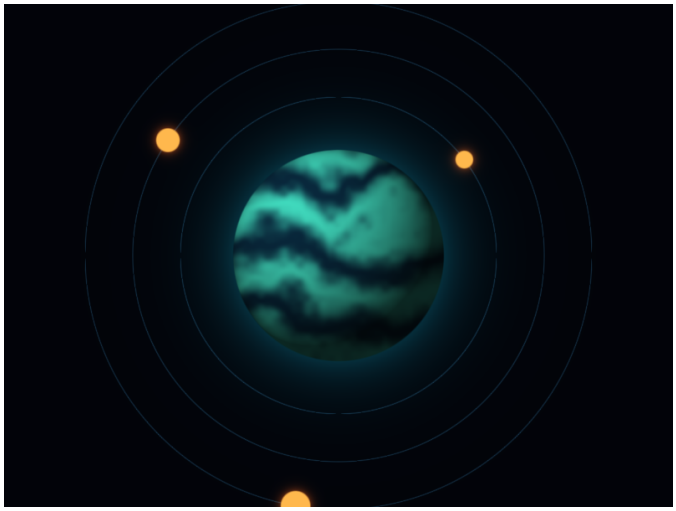
IMPLEMENTATION AND MEASURED EVIDENCE

# Lab 07: Shaders

**Nathawat Wattanarapeepong** | Computer Graphics, section 21
Prepared with Codex (GPT-6). Measurements taken on this Mac on 2026-10-09.



Part II: deterministic terrain, six octaves.



Part V: original Tidal Observatory, t = 1 s.

## Completion and provenance

Only the six permitted shader files were edited in the repository. All four supplied reference comparisons report **MATCH, 0.00% differ**. The final toy passes all source checks and the 2.0 ms budget. This report contains AI-assisted explanations and actual local measurements; it does not claim that the student independently performed or predicted the experiments.

Frames are direct 800 x 600 GPU readbacks. Panel images are exact stdout transcripts rendered for legibility, not desktop screenshots. The original text logs and temporary comparison shaders are included in evidence/. Terminal screenshot access was unavailable through the automation tool.

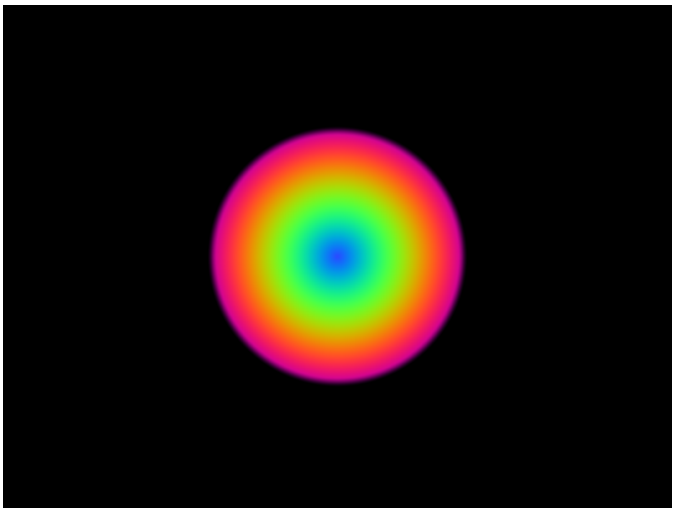
```
info | captured stdout

  device          Apple M2 Pro
  api             1.3.357
  subgroup        32 lanes
  timestamps      1 ns per tick
  statistics      pipeline statistics unsupported
  vertex inputs   31 attributes max, and Part I uses none of them
  glslc           /usr/local/bin/glslc
  version         shaderc v2026.3 v2026.3
  shaders         /tmp/lab07-work/source/shaders
  circle.frag     TASK 1 done
  noise.glsl      TASK 2 done
  sdf.frag        TASK 3 done
  heightmap.vert  TASK 4 done
  heightmap.frag  TASK 4 done
  mytoy.frag      TASK 5 done
```

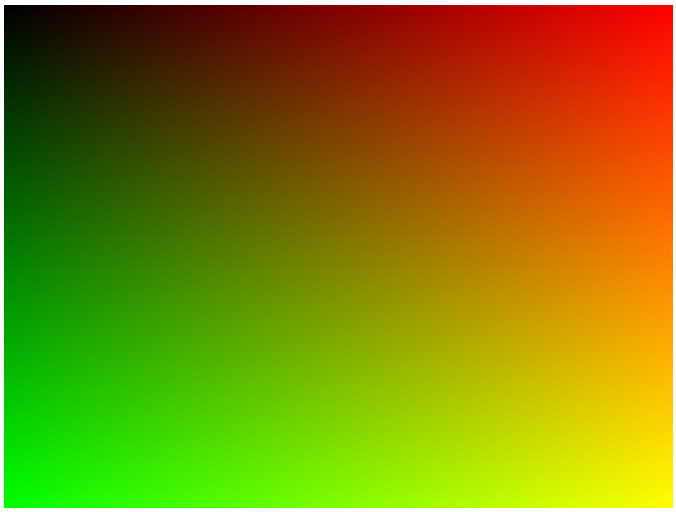
Machine panel from the built info executable (equivalent to the executable run by make info). All six TASK lines are done.

Method: Release build, Vulkan SDK 1.4.357.1, MoltenVK, no concurrent lab benchmarks. The provided timer uses 5 warmups and 21 samples, each with 20 repeated passes; sweeps use 20 warmups and 200 passes per sample. Timing is the GPU pass time, not whole application frame time. Pipeline statistics are unsupported, so invocation counts below are source-level estimates.

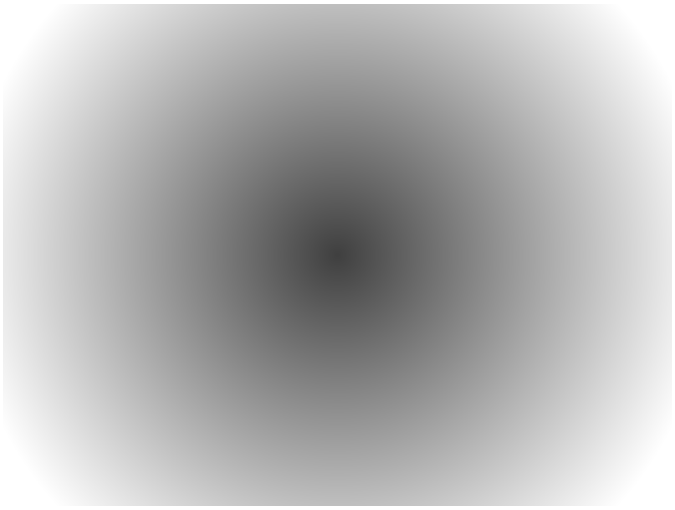
# Part I: the pixel is a function



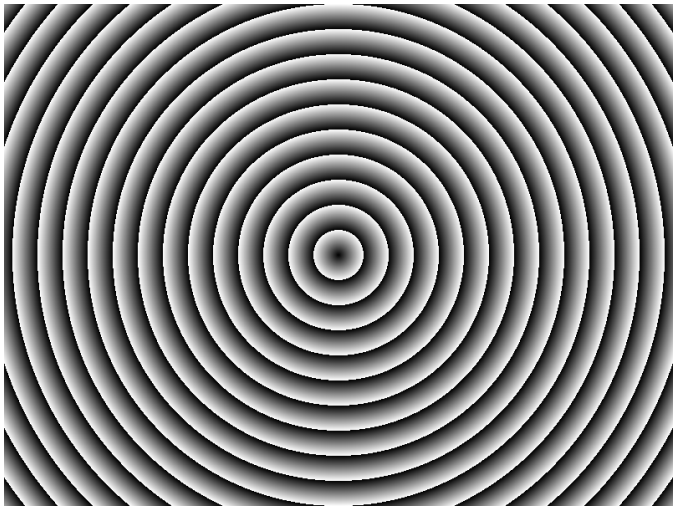
Mode 0: animated disc at t = 1.



Mode 1: UV. Green is bottom left.



Mode 2: signed distance mapped to grey.



Mode 3: evenly spaced distance contours.

## 1.1 Coordinates and the Y flip

With this positive-height Vulkan viewport, gl_FragCoord.y increases downwards. The mapping p = (2*fragCoord - resolution)/resolution.y already makes the top negative Y; the old viewportTransform() flip is unnecessary for this pixel-space shader. The green corner is bottom left, pixel (0,599). To port a raw ShaderToy shader, use fragCoord.y = resolution.y - gl_FragCoord.y. In Part V, shadertoy.glsl already performs that line, so mainImage must not flip again.

## 1.2 Edge width in pixels

A p unit spans R.y/2 pixels. The full -0.01 to +0.01 transition therefore spans 0.01*R.y: **6 px at 800 x 600**, and **21.6 px at 3840 x 2160**. For w = fwidth(d), the full -w to +w transition is approximately **2 px on an axis-aligned rim**, up to 2*sqrt(2) = 2.83 px on a diagonal, at either resolution. The handout calls it one pixel; that describes a derivative-sized half-width, not the full transition.

## 1.3 How a fragment obtains derivatives

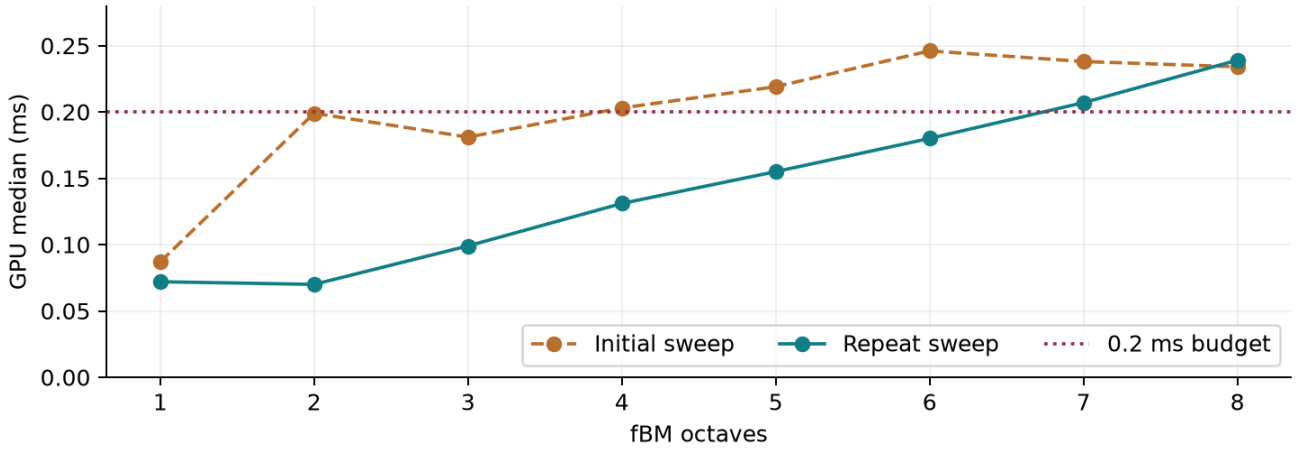
fwidth(d) = abs(dFdx(d)) + abs(dFdy(d)). The GPU compares values across a 2 x 2 fragment quad, including helper invocations at primitive edges. Lab 06 Part III introduced neighboring-lane execution; a subgroup is larger than a quad and is not the derivative unit. Derivatives must be evaluated with uniform control flow across the quad. [2]

# Part II: what an octave costs

## 2.1 Prediction recorded before the first run

Predicted T(n) = a + b*n, b > 0: roughly linear, with T(8)/T(1) below 8 because fixed pass costs remain. Pure noise arithmetic alone predicts 8. This was Codex's pre-run prediction, saved with a timestamp in predictions-before-measuring.txt; it is not presented as the student's unrecorded prior thought.

| Octaves | Initial median (ms) | Repeat median (ms) |
|---|---|---|
| 1 | 0.087 | 0.072 |
| 2 | 0.199 | 0.070 |
| 3 | 0.181 | 0.099 |
| 4 | 0.203 | 0.131 |
| 5 | 0.219 | 0.155 |
| 6 | 0.246 | 0.180 |
| 7 | 0.238 | 0.207 |
| 8 | 0.234 | 0.239 |




## 2.2 Slope, ratio, and nonlinearity

Initial least-squares slope: **0.01708 ms/octave**, intercept 0.1240 ms, R-squared 0.673; 8/1 ratio **2.690**. The non-monotonic values prompted a repeat, retained alongside the first run. Repeat slope: **0.02525 ms/octave**, intercept 0.03050 ms, R-squared 0.982; ratio **3.319**. The repeat supports the predicted near-linear trend. These passes also measure rasterization, band coloring, render-pass overhead, clock behavior and contention; the timer cannot isolate noise arithmetic. Background load or power-state changes are plausible causes, not measured diagnoses.

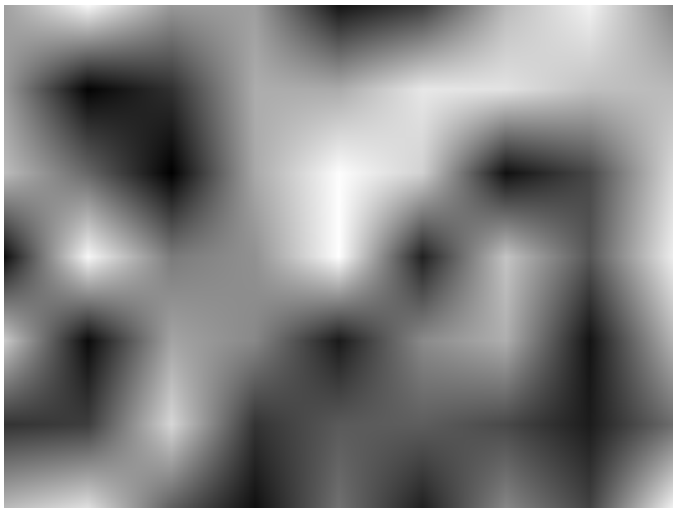
## 2.3 A 0.2 ms budget

The repeat admits **6 octaves** (0.180 ms; 7 costs 0.207). The initial sweep admits only **3** (0.181; 4 costs 0.203), so six is not a guarantee under all observed conditions. At 4K, 3840*2160/(800*600) = **17.28**: even the repeat's one-octave cost becomes 1.244 ms, so **no tested positive octave count** fits. This extrapolates constant cost per pixel and unchanged clocks/occupancy; no 4K GPU run was performed.

# Part II: reproducible, smooth noise



Hermite interpolation, mode 2.



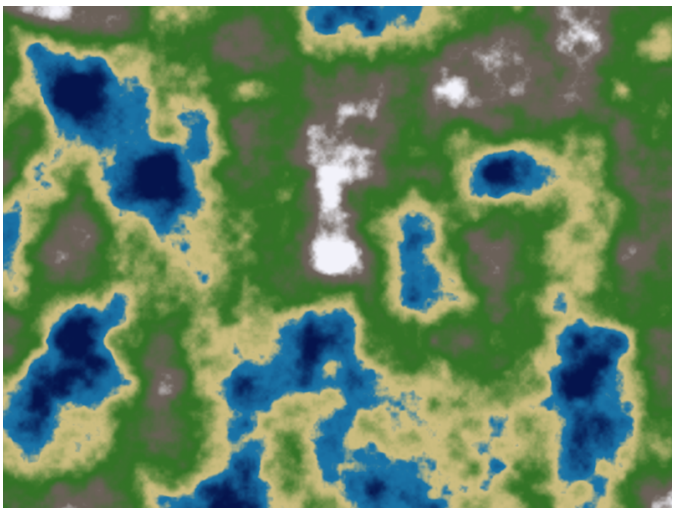
Temporary linear interpolation, same lattice.

## 2.4 Why the hash matters

The sine hash multiplies a transcendental result by 43758.5453 and takes fract, amplifying small implementation-dependent floating-point differences into different random-looking values. The supplied PCG uses defined unsigned 32-bit multiply, shift and xor, so the integer hash is reproducible. The final float conversion/interpolation can still round; the panel checks the actual 8-bit result. Our lattice probe reads **169**, matching the CPU; one octave reads **84**; the eight-octave maximum is **0.914**, below 1 - 0.5^8 = 0.99609375.

## 2.5 Why Hermite instead of linear weights

For H(f) = f*f*(3 - 2*f), H'(f) = 6*f*(1-f), which vanishes at both cell boundaries. Adjacent cells meet with a smooth first derivative. With plain f, the slopes generally jump across a lattice boundary. In the saved linear mode-2 frame, the 100-pixel cell grid has visible straight ridges and angular, diamond-like transitions; the Hermite frame has rounded transitions. Values are still continuous in the linear version; it is the slope that jumps. The temporary variant was restored after capture.



Final terrain: six-octave image matches reference.

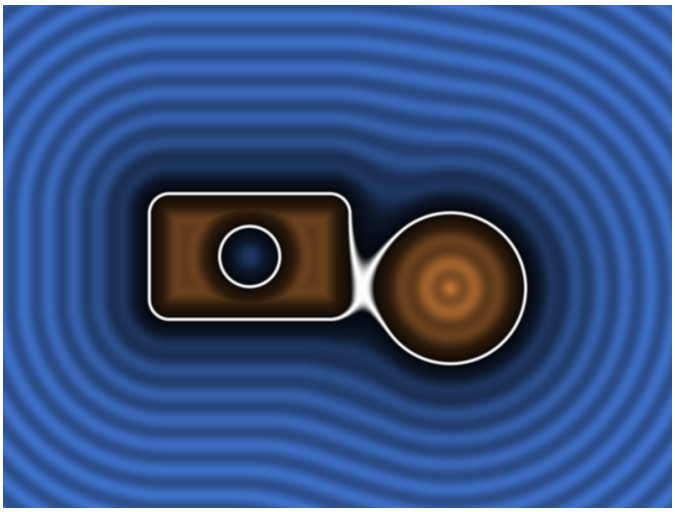


ShaderToy no-texture default copied as a temporary test.

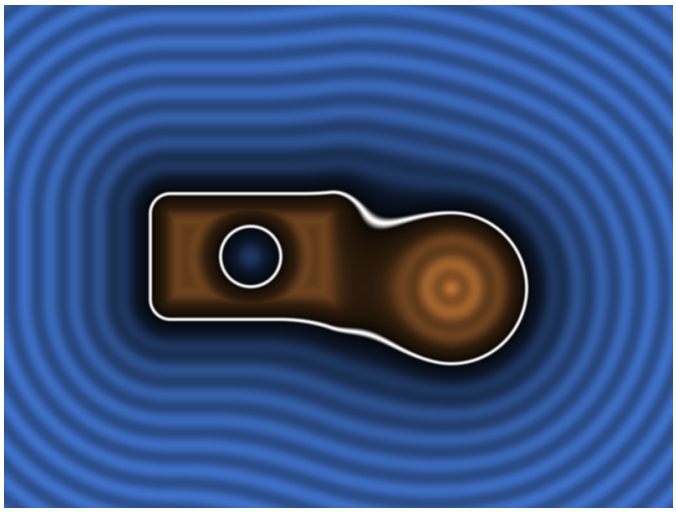
## Wrapper portability check

The mainImage shown at shadertoy.com/new was copied into an isolated temporary mytoy.frag and rendered locally. Its UV cosine gradient ran successfully, then the original planet shader was restored. Pixel values were checked against the bottom-left-origin equation. This was a separate compiled check, not a claim of completing a live editor copy/paste session. [3]

# Part III: signed distance and blending



k = 0.20: specified final blend.



k = 0.40: wider, connected fillet.

## 3.1 Union and subtraction

**Union:** min(d1,d2) is negative wherever either primitive contains the point, so its zero boundary encloses their union (the combined field need not be exact everywhere). **Subtraction:** max(d1,-d2) is negative only inside shape 1 and outside shape 2, which removes shape 2 and reverses the sign at its boundary.

## 3.2 What k changes

The blend weight h is clamped from 0.5 + 0.5*(d2-d1)/k. Within |d2-d1| < k, subtracting k*h*(1-h) lowers the field and rounds the junction. At k = 0.40 the waist visibly fills and its contours bend farther out; coverage rises from **12.5% to 13.6%**. Ship **0.20** to preserve the prescribed scene and reference match. The variant's blend probe differs from the fixed CPU scene by design; that is not a failure of the restored solution.

| Probe pixel | Shader d | CPU d | Classification |
|---|---|---|---|
| (220,300) | -0.129 | -0.128 | Inside body |
| (60,60) | 0.702 | 0.700 | Outside |
| (430,300) | 0.012 | 0.010 | Blend / outside |
| (295,300) | 0.114 | 0.118 | Hole / outside |

## 3.3 A bound is not an exact distance

The panel's unquantized blended field is **0.010**, versus **0.052** to the nearer original primitive: 0.042 smaller, about 81% relative to that primitive distance. The quantized shader readback is 0.012, with a channel step of 2/255 = 0.00784. The comparison alone does not establish the exact distance to the newly blended zero set, which is a different surface.

An algorithm that moves a point to the nearest surface in one step using p -= d*normalize(gradient(d)) will under-project if it assumes this bound is an exact distance. Conservative sphere tracing does **not** break merely because the positive field underestimates distance: it takes smaller safe steps. The handout's primitive-distance comparison should not be described as proof of the new surface's exact distance.

# Part IV: noise in the vertex stage

## 4.1 Prediction and the handout counting error

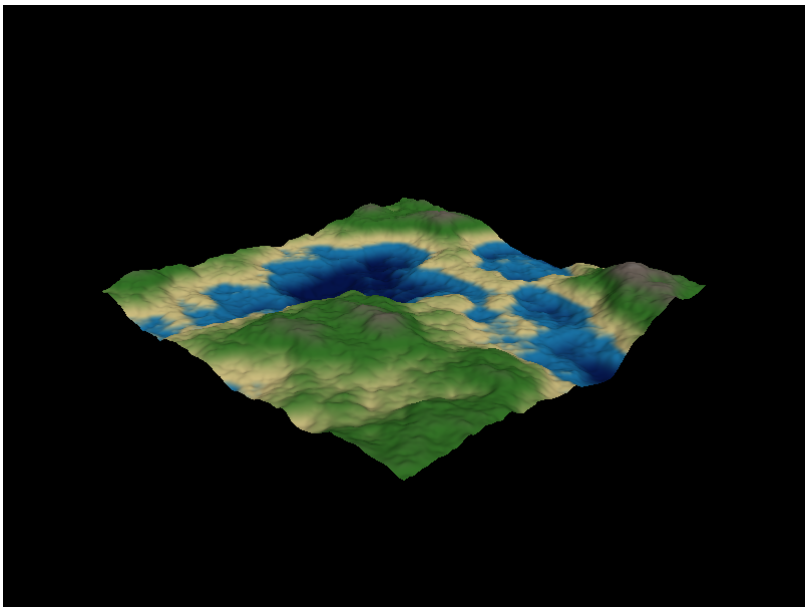
Before measuring, the predicted ordering was vertex N64 < vertex N128 < fragment N64 < vertex N256 < vertex N512. The specified central differences need **five** height() calls: one for position and two neighbors on each of two axes. Two subtractions are not two height evaluations. The implementation follows the displayed formula, including e = 2/N, and matches the reference.

| Row | Vertices / pixels | fBM calls (estimate) | Median (ms) |
|---|---|---|---|
| Vertex N64 | 4,096 vertices | 20,480 | 0.041 |
| Vertex N128 | 16,384 vertices | 81,920 | 0.087 |
| Fragment N64 | 90,772 covered pixels | 90,772 | 0.096 |
| Vertex N256 | 65,536 vertices | 327,680 | 0.314 |
| Vertex N512 | 262,144 vertices | 1,310,720 | 1.041 |

## 4.2 Measured ordering and scaling

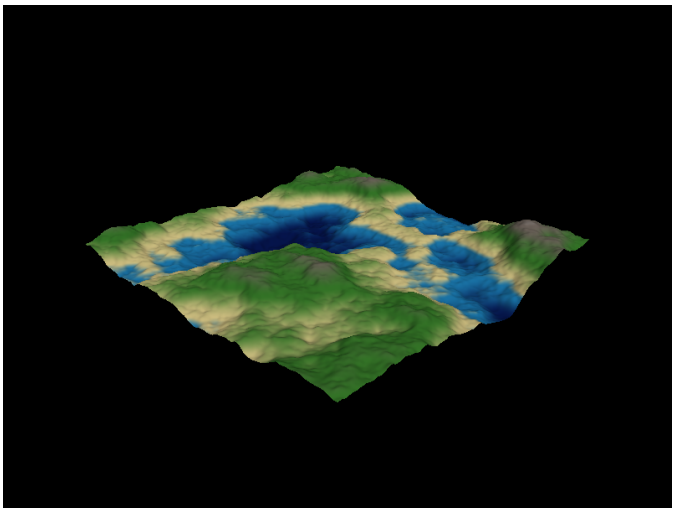
The initial five rows match the prediction. Using the correct five-call formula gives 90,772/(5*4,096) = **4.432** times as many source fBM calls in the fragment comparison, while the measured time ratio is 0.096/0.041 = **2.341**. The handout's three-call assumption would give 7.387, but does not describe the supplied formula. Counts alone predict direction, not an exact time ratio: fixed costs, execution efficiency, vertex reuse and rasterization also contribute.

Doubling N multiplies source vertex work by four, while the measured vertex ratios are **2.12, 3.61, 3.32**. Pixel shading and fixed work do not quadruple, and the grid develops small triangles. Covered pixels also exclude helper invocations and any overdraw; vertex cache behavior can cause repeated vertex processing. These are estimates, not hardware counters. In a later coarse-grid run, the near-tied N128 and fragment rows reverse (0.089 vs 0.088 ms), showing why their order should not be generalized.

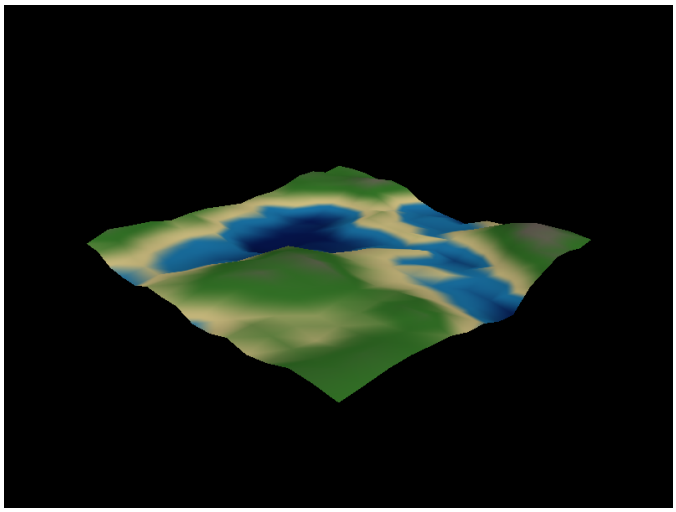


N = 256: center normal (149,244,81), 65,536 vertices, 390,150 indices, 130,050 triangles; 86,734 nonblack pixels (18.1%); reference MATCH.

# Part IV: normals and undersampling



N = 256: detailed slopes and silhouette.



N = 16: broad triangles lose fine relief.

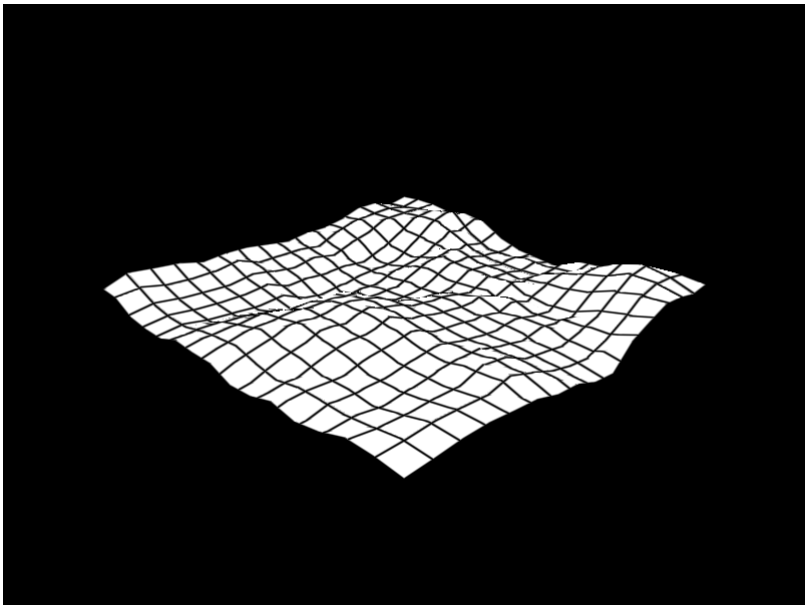
## 4.3 Why the normal needs neighboring heights

For the graph y = h(x,z), a normal is proportional to (-dh/dx, 1, -dh/dz). Central differences approximate dh/dx with [h(x+e,z)-h(x-e,z)]/(2e), and similarly for z. Multiplying the vector by 2e gives the exact expression used in heightmap.vert. This needs four neighboring samples plus the central position sample: **five total**. A three-call version would require forward differences, which is a different method.

normalize(cross(dFdx(inWorld),dFdy(inWorld))) in the fragment stage would give the normal of the rendered triangle plane, with cross-product order adjusted for orientation. It would show planar facets and lighting discontinuities across triangle boundaries on a coarse mesh, rather than the interpolated smooth finite-difference normals. The geometric outline would still be set by the vertices.

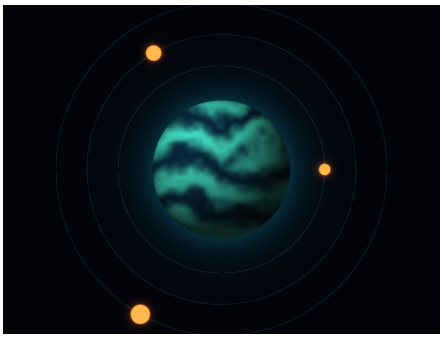
## 4.4 The silhouette at a coarse grid

At N = 16, the saved final and grid views show long straight outline segments and flattened small peaks. The highest of six noise octaves has nominal lattice frequency 1.5*2^5 = 48 cells per world unit; the grid samples every 2/(N-1), far too sparsely to represent it. Fine terrain is lost or folds into false coarser structure: **aliasing / undersampling**, the same failure behind texture shimmer and moire. These captures use --grid 16, the command-line equivalent of repeated [ presses.

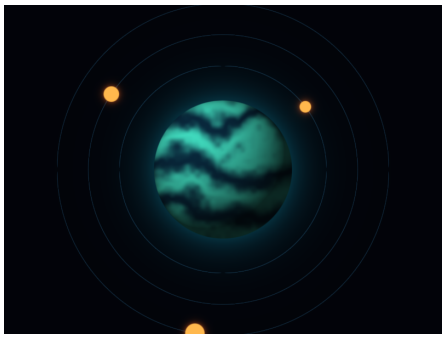


Mode 2 at N = 16: the coarse cells are directly visible. The host places vertices 2/(N-1) apart; the assignment's requested finite-difference step is 2/N, which is slightly smaller than a mesh cell.

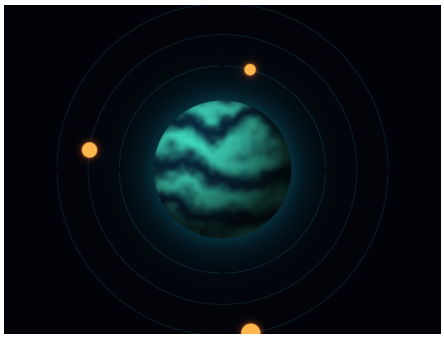
# Part V: Tidal Observatory



t = 0 seconds



t = 1 second



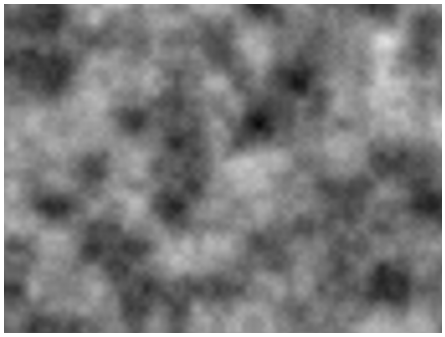
t = 2 seconds

## 5.1 How the terms make the picture

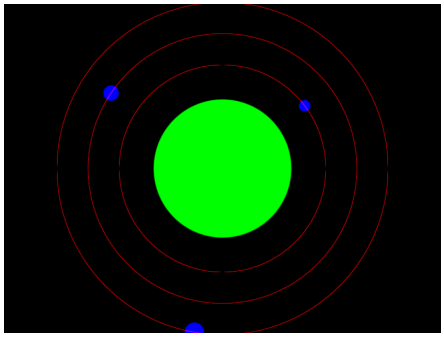
**q** is the centered, aspect-correct pixel position relative to **center**; pressing the mouse moves center to the pointer. **planet = length(q) - PLANET_RADIUS** is the Part III circle SDF; **planetMask** uses its derivative width to antialias the edge. **clouds** is Part II fBM sampled at 4*q plus a time-dependent translation. **bands = 0.5 + 0.5*sin(24*q.y + 9*clouds + 0.4*iTime)** bends moving latitude bands with noise, and **surface** mixes dark teal and mint. **z** reconstructs a hemisphere for **normal**; **light** applies ambient plus Lambert shading. **halo = exp(-9*abs(planet))** adds atmospheric glow. Each **track** is an absolute circle SDF with a 0.0015 half-width; **moonCenter** follows (cos(angle),sin(angle)) at radius 0.63 + 0.19*j. The **angle** increases with iTime, with slower outer moons; **moonMask** adds warm discs and an exponential glow.



Mode 1: circle-union field.



Mode 2: cloud fBM.



Mode 3: tracks / planet / moons in RGB.

## Interaction and wrapper checks

The wrapper supplies bottom-left-origin fragCoord and iMouse; no second Y flip is applied. A separate temporary host harness injected mouse = (600,150,1,0) in Vulkan coordinates while rendering the exact final shader. Its mode-3 green disc moved to that pixel region, confirming the uniform-to-picture mapping. Physical dragging still needs the student's own brief window check; the native automation could not bind the bare GLFW executable. Repository host code was not changed.

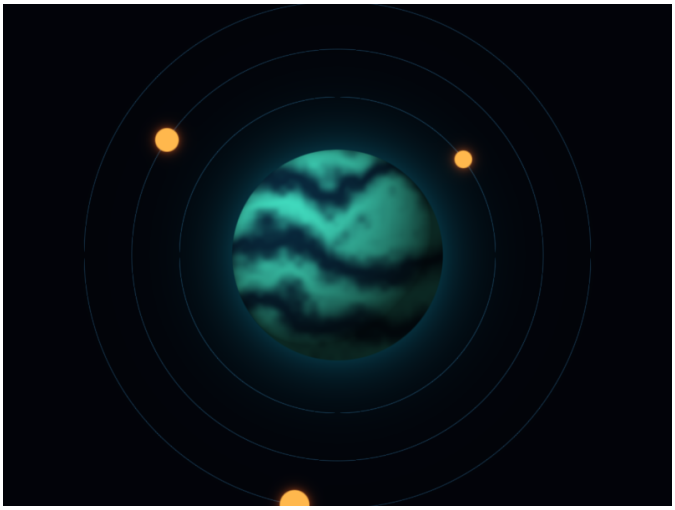
All five rule checks pass: time animation, mouse response in the shader, an sdf_circle primitive, fBM, distinct iMode views, and a median well below 2.0 ms. The source scanner reports four names; the saved images and timings provide the behavioral evidence beyond those names.

# Part V: cost and design tradeoffs

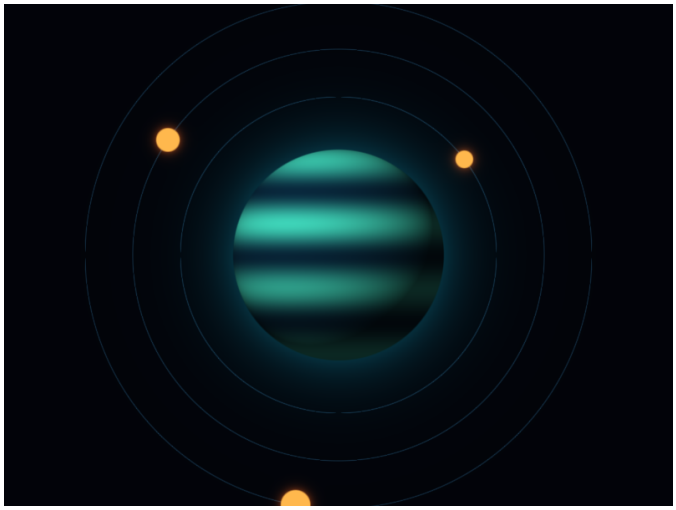
| Controlled variant, 800 x 600 | Median (ms) | p95 (ms) |
|---|---|---|
| Eight cloud octaves, --octaves 8 | 0.253 | 0.387 |
| Final four-octave cap (toy-final log) | 0.142 | 0.292 |
| Cloud fBM removed; clouds = 0.5 | 0.044 | 0.126 |

## 5.2 Cost of the most expensive term

Replacing the complete cloud fBM expression with the constant 0.5 reduced the median from **0.142 to 0.044 ms**. The marginal saving is **0.098 ms, or 69.0%** of the final measured pass. It includes compiler simplification downstream of the constant, so it is not a hardware instruction-level attribution. Other unmodified final runs were 0.135 and 0.138 ms; the repeat variation does not change the budget result.



Final: cloud-distorted bands.



Ablation: regular smooth latitude stripes.

## 5.3 One budget-oriented change

The tested high-detail variant used eight octaves. The final shader caps CLOUD_OCTAVES at four, reducing the measured median by **0.111 ms (43.9%)**. The large cloud shapes remain, while the smallest wrinkles and fine surface grain disappear. Eight octaves already passed 2.0 ms; this was a deliberate margin improvement, not a claim that a failing shader needed rescue. The final 0.142 ms is 7.1% of the allowed budget.

## Constants to explain in the live exercise

| Change before saving | Prediction |
|---|---|
| PLANET_RADIUS: 0.42 to 0.50 | Radius grows from 126 to 150 px at 600 px height; orbit radii stay fixed. |
| ORBIT_SPEED: 0.65 to 1.30 | Each moon's angular speed doubles; each orbital period halves. |
| CLOUD_OCTAVES: 4u to 2u | Two finest noise layers disappear; the fBM ceiling falls from 0.9375 to 0.75. |
| Band frequency: 24.0 to 12.0 | Half as many latitude oscillations across the same vertical distance. |

# AI use, verification, and reproduction

## Two requests: user-revised summary

The following wording was supplied by the user when revising this report. It summarizes the requested assistance; it is not a verbatim transcript of the original implementation session.

**1. give me a step by step to do this project**
**Model:** Codex, GPT-6.
**Idea:** break the assignment into an ordered workflow for implementation, checking results, and submission.
**Verification:** the workflow was checked against the handout, the six permitted shader files, the provided CPU probes, and the local build and render results.

**2. do the document for me**
**Model:** Codex, GPT-6.
**Idea:** organize the measured results, shader images, five write-ups, and AI-use record into a report.
**Verification:** reported numbers were checked against captured logs; the PDF was rendered and visually reviewed for layout and legibility.

**Session record:** the original assignment prompt was "/Users/santa.ntw/Downloads/instructions-1032980-17897008458766.pdf do this assignment for me". Codex implemented the shaders and collected the evidence as well as writing the report. No sub-agents were used.
**Prior knowledge:** the student's starting understanding of GLSL, SDFs and noise was not provided and must still be personalized.

**Verification actually performed by Codex:** read all task requirements and host probes; recorded predictions before the first completed run; compiled all shaders and executables; compared the four GPU frames to supplied references; checked circle center/corners, the integer hash lattice probe, one-octave amplitude, ceiling, four SDF probes, and heightmap normals; captured the requested variants and time frames; measured an fBM ablation; read the shader equations; checked mouse response using an isolated uniform-injection harness. The report does not attribute these actions to the student.

## Reproduce without editing supplied build files

The repository path contains an apostrophe and spaces, exposing a quoting defect in the provided CMake shader commands. A temporary symlink /tmp/lab07-work/source points to the original repository. The successful build used cmake -S /tmp/lab07-work/source -B /tmp/lab07-work/build-safe -DCMAKE_BUILD_TYPE=Release with the installed Vulkan include/library paths. The complete reusable commands are in RUN-INSTRUCTIONS.txt.

Each panel uses the supplied executable with --headless at time 1; --save captures its measured image. Comparison variants are kept outside the repository and compiled into temporary SPIR-V, then the final shader is restored. During these variants the panel's source-name scanner still reads the repository source; only the rendering and timing lines should be used as variant evidence. Reference comparisons intentionally differ for debug modes or altered constants.

## Readiness and sources

Before submission: personalize the prior-knowledge field, physically drag in the toy window, and be prepared to explain one live constant change. The code and machine results are complete; personal understanding cannot be verified by an assistant. If the instructor strictly requires desktop panel screenshots, replace the labelled stdout images with terminal screenshots of the saved logs or fresh runs.

[1] CP479, Lab 07: Shaders, supplied 16-page handout, 2026-09-18. The included skeleton and reference images define the check values.
[2] Khronos, GLSL Built-In Functions: docs.vulkan.org/glsl/latest/chapters/builtinfunctions.html. In particular, reversed smoothstep edges are undefined; the implementation uses 1 - smoothstep(-w,w,d), preserving the intended ramp. Derivative behavior and terminology were checked here.
[3] ShaderToy default no-texture mainImage: shadertoy.com/new, viewed in Chrome during this task. Used only for the temporary wrapper test.

# Captured panels: Parts I to III

```
circle | captured stdout
  shaders        fullscreen.vert + circle.frag, no vertex buffer
  uniform        176 bytes, time 1.000 s, mode 0
  coverage       15.9% of the target, pi*0.25 / (2 * 8/3) = 14.7% before the AA ring
  centre         pixel (400, 300) reads (42, 77, 255), cpu says (42, 77, 255)
  mode 1         (0, 0) reads (0, 0, 0), expected (0, 0, 0)
  mode 1         (799, 0) reads (255, 0, 0), expected (255, 0, 0)
  mode 1         (0, 599) reads (0, 255, 0), expected (0, 255, 0)
  image          MATCH (0.00% differ)
  gpu time       0.048 ms median, 0.053 ms p95
```

Part I: circle --headless. Center, corners and image match.

```
terrain | captured stdout
  shaders        fullscreen.vert + terrain.frag, which includes noise.glsl
  uniform        time 1.000 s, mode 0, octaves 6
  lattice        pixel (300, 200) is p = (3, 2): value_noise reads 169, cpu hash(uvec2(1003, 1002)) is 169
  one octave     fbm(p, 1) reads 84, half of 169 is 84
  fbm ceiling    1 - 0.5^8 = 0.996, brightest pixel of the image 0.914
  octaves 1        0.087 ms median
  octaves 2        0.199 ms median
  octaves 3        0.181 ms median
  octaves 4        0.203 ms median
  octaves 5        0.219 ms median
  octaves 6        0.246 ms median
  octaves 7        0.238 ms median
  octaves 8        0.234 ms median
  image          MATCH (0.00% differ) at 6 octaves
  gpu time       0.704 ms median, 0.883 ms p95
```

Part II: initial terrain --headless. Non-monotonic timing values are preserved.

```
sdf | captured stdout
  shaders        fullscreen.vert + sdf.frag
  uniform        time 1.000 s, mode 0
  probe          (220, 300) inside the body     shader d -0.129  cpu d -0.128  inside
  probe          ( 60,  60) outside everything  shader d  0.702  cpu d  0.700  outside
  probe          (430, 300) in the blend        shader d  0.012  cpu d  0.010  outside
  probe          (295, 300) inside the hole     shader d  0.114  cpu d  0.118  outside
  bound          in the blend the field says 0.010, the nearer primitive is
                 0.052 away: after a smooth union d is a bound, not a distance
  coverage       12.5% of the target is shape, the rest is exactly black
  image          MATCH (0.00% differ)
  gpu time       0.031 ms median, 0.239 ms p95
```

Part III: sdf --headless. Quantized shader probes agree within one distance-channel step.

# Captured panels: Part IV and V

```
heightmap | captured stdout
  shaders        heightmap.vert + heightmap.frag, the vertex stage does fbm
  grid           N 256: N*N = 65536 vertices, 6*(N-1)^2 = 390150 indices, 130050 triangles
  camera         eye (2.2, 1.6, 2.2), 45 deg, reversed-Z: clear 0, compare GREATER
  normals        mode 1 centre pixel (149, 244, 81), flat ground would read (128, 255, 128)
  coverage       18.1% of the target, 86734 fragments
  vertex stage   N  64     0.041 ms median,    4096 vertices
  vertex stage   N 128     0.087 ms median,   16384 vertices
  vertex stage   N 256     0.314 ms median,   65536 vertices
  vertex stage   N 512     1.041 ms median,  262144 vertices
  fragment stage N  64     0.096 ms median, 90772 fragments run the same fbm
  image          MATCH (0.00% differ) at N 256
  gpu time       0.246 ms median, 0.385 ms p95
```

Part IV: heightmap --headless, default N = 256, six octaves.

```
toy-final | captured stdout
  shaders        fullscreen.vert + mytoy.frag, through shadertoy.glsl
  uniform        time 1.000 s, mode 0, octaves 5
  gpu time       0.142 ms median, 0.292 ms p95
  budget         2.000 ms at 800x600: PASS
  uses noise.glsl  yes
  uses sdf_        yes
  uses iMouse      yes
  uses iMode       yes
  TODO removed     yes
  coverage       100.0% of the target is not black
  frames         /tmp/lab07-work/source/report/toy_0.png, toy_1.png, toy_2.png
```

Part V: final source with CLOUD_OCTAVES = 4; host octaves = 5, so the shader evaluates min(5,4).

## Part I check arithmetic

The disc occupies pi*0.5^2 / (2*8/3) = 14.726% before the antialias ring; measured nonblack coverage is 15.9%. At pixel (400,300), p = (1/600,1/600), d = -0.497643, and the t = 1 cosine palette gives (42,77,255) after UNORM rounding. No sRGB conversion is applied.

## Scope of image matching

The supplied comparator marks a pixel different only if any RGB channel differs by more than 3 levels, and accepts an image with fewer than 1% such pixels. Thus 0.00% differ means agreement under that tolerance, not a promise of bit-identical floating-point evaluation on every GPU. All source changes remain confined to the six permitted shader files.

# Additional measurement evidence

```
terrain-repeat | captured stdout
  shaders        fullscreen.vert + terrain.frag, which includes noise.glsl
  uniform        time 1.000 s, mode 0, octaves 6
  lattice        pixel (300, 200) is p = (3, 2): value_noise reads 169, cpu hash(uvec2(1003, 1002)) is 169
  one octave     fbm(p, 1) reads 84, half of 169 is 84
  fbm ceiling    1 - 0.5^8 = 0.996, brightest pixel of the image 0.914
  octaves 1         0.072 ms median
  octaves 2         0.070 ms median
  octaves 3         0.099 ms median
  octaves 4         0.131 ms median
  octaves 5         0.155 ms median
  octaves 6         0.180 ms median
  octaves 7         0.207 ms median
  octaves 8         0.239 ms median
  image          MATCH (0.00% differ) at 6 octaves
  gpu time       0.153 ms median, 0.311 ms p95
```

Part II repeat: final shader, default mode, all lattice/image checks preserved.

```
toy-eight-octaves | captured stdout
  shaders        fullscreen.vert + mytoy.frag, through shadertoy.glsl
  uniform        time 1.000 s, mode 0, octaves 8
  gpu time       0.253 ms median, 0.387 ms p95
  budget         2.000 ms at 800x600: PASS
  uses noise.glsl  yes
  uses sdf_        yes
  uses iMouse      yes
  uses iMode       yes
  TODO removed     yes
  coverage       100.0% of the target is not black
  frames         /tmp/lab07-work/source/report/toy_0.png, toy_1.png, toy_2.png
```

Eight-octave variant: extra fine detail; 0.253 ms median.

```
toy-no-clouds | captured stdout
  shaders        fullscreen.vert + mytoy.frag, through shadertoy.glsl
  uniform        time 1.000 s, mode 0, octaves 5
  gpu time       0.044 ms median, 0.126 ms p95
  budget         2.000 ms at 800x600: PASS
  uses noise.glsl  yes
  uses sdf_        yes
  uses iMouse      yes
  uses iMode       yes
  TODO removed     yes
  coverage       100.0% of the target is not black
  frames         /tmp/lab07-work/source/report/toy_0.png, toy_1.png, toy_2.png
```

Cloud fBM ablation: 0.044 ms median. The source scanner reads the final repository source, not the temporary binary variant.